Nonlinear equations in one variable and systems of equations in two variables — Part 2

Advanced Math · Math · 82 questions

2.1 034071-DC · Medium

$(x-4)(x+2)(x-1)=0$

What is the product of the solutions to the given equation?

A. 8

B. 3

C. -3

D. -8

2.2 027107-DC · Medium

$6x^2+5x-7=0$

What are the solutions to the given equation?

A. $\frac{-5\pm\sqrt{25+168}}{12}$

B. $\frac{-6\pm\sqrt{25+168}}{12}$

C. $\frac{-5\pm\sqrt{36-168}}{12}$

D. $\frac{-6\pm\sqrt{36-168}}{12}$

2.3 070620-DC · Medium

$x^2-ax+12=0$

In the equation above, a is a constant and $a>0$. If the equation has two integer solutions, what is a possible value of a ?

Student-produced response: _____

2.4 040b3ec1-691d-42ae-92e9-096acde7ab60 · Medium

$$w^2 + 12w - 40 = 0$$

Which of the following is a solution to the given equation?

A. $6 - 2\sqrt{19}$

B. $2\sqrt{19}$

C. $\sqrt{19}$

D. $-6 + 2\sqrt{19}$

2.5 4f8dda48-f98a-469b-b065-7f377d7390ab · Medium

The equation $12t + b = c$ relates the variables t, b, and c. Which of the following correctly expresses the value of $c - b$ in terms of t?

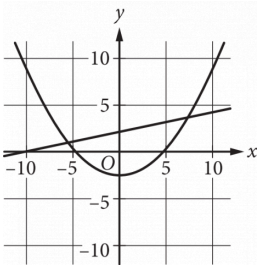
A. $\frac{t}{12}$

B. t

C. $t + \frac{1}{12}$

D. $12t$

2.6 026003-DC · Medium



A system of equations consists of a quadratic equation and a linear equation. The equations in this system are graphed in the xy-plane above. How many solutions does this system have?

A. 0

B. 1

C. 2

D. 3

2.7 88cb70a7-15a5-452e-9010-61cf2f4f5a6f · Medium

$$x^2 - 12x + 27 = 0$$

How many distinct real solutions does the given equation have?

A. Exactly two

B. Exactly one

C. Zero

D. Infinitely many

2.8 025700-DC · Medium

$$\frac{2(x+1)}{x+5} = 1 - \frac{1}{x+5}$$

What is the solution to the equation above?

A. 0

B. 2

C. 3

D. 5

2.9 a1e6d838-3642-4546-9031-e113b254b3f2 · Medium

$$x = 8ab + 9$$

The given equation relates the positive numbers a, b, and x. Which equation correctly expresses a in terms of b and x?

A. $a = \frac{x}{8} - b + 9$

B. $a = \frac{x}{8b+9}$

C. $a = \frac{8b+9}{x}$

D. $a = 8xb + 9$

2.10 f70845bb-9302-4075-b020-3ece6ab22306 · Medium

$$\frac{55}{x+6} = x$$

What is the positive solution to the given equation?

Student-produced response: _____

2.11 015183-DC · Medium

Blood volume, V_B, in a human can be determined using the equation $V_B = \frac{V_P}{1-H}$, where V_P is the plasma volume and H is the hematocrit (the fraction of blood volume that is red blood cells). Which of the following correctly expresses the hematocrit in terms of the blood volume and the plasma volume?

A. $H = 1 - \frac{V_P}{V_B}$

B. $H = \frac{V_B}{V_P}$

C. $H = 1 + \frac{V_B}{V_P}$

D. $H = V_B - V_P$

2.12 08148-DC · Medium

$$y = x + 1$$
$$y = x^2 + x$$

If (x,y) is a solution to the system of equations above, which of the following could be the value of x ?

A. –1

B. 0

C. 2

D. 3

2.13 df05bbf9-4751-4ac6-b485-8df58c021387 · Medium

If $4x - 4 = 112$, what is the positive value of $x - 1$?

Student-produced response: _____

2.14 692774ab-373f-4544-a043-aea38e8b5f3f · Medium

$$7x^2 - 20x - 32 = 0$$

What is the positive solution to the given equation?

Student-produced response: _____

2.15 08453-DC · Medium

$y = ax^2 - c$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy-plane?

A. Zero

B. One

C. Two

D. More than two

2.16 02431d60-e74e-4725-8ae2-b4a8af3e1ddd · Medium

$$3xx - 4x + 5 = 0$$

What is one of the solutions to the given equation?

A. -4

B. 0

C. 3

D. 5

2.17 da10ca3e-b5b5-46f6-be6d-751c5687b72a · Medium

$$52x^3 + 64x^4 - 81 = 0$$

How many distinct real solutions does the given equation have?

A. Exactly two

B. Exactly three

C. Exactly five

D. Exactly seven

2.18 2b61c753-bb12-4153-963a-865282e1a793 · Medium

$$b = 42cf$$

The given equation relates the positive numbers b, c, and f. Which equation correctly expresses c in terms of b and f?

A. $c = \frac{b}{42f}$

B. $c = \frac{b - 42}{f}$

C. $c = 42bf$

D. $c = 42 - b - f$

2.19 070618-DC · Medium

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

Student-produced response: _____

2.20 e3cbd1d8-7084-48db-a4b9-384e8121846d · Medium

$$7m = 5n + p$$

The given equation relates the positive numbers m, n, and p. Which equation correctly gives n in terms of m and p?

A. $n = \frac{5p}{7m}$

B. $n = \frac{7m}{5} - p$

C. $n = 57m + p$

D. $n = 7m - 5 - p$

2.21 a5f0747c-79a9-4cc0-a9eb-aa9d68a2d5e8 · Hard

$$-9x^2 + 30x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c?

A. 3

B. 0

C. -25

D. -53

2.22 cc8d5c5d-bc43-4311-981f-7eb0259aaf7a · Hard

$$-x^2 + bx - 676 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b?

Student-produced response: _____

2.23 09411-DC · Hard

$$y = x^2 + 3x - 7$$
$$y - 5x + 8 = 0$$

How many solutions are there to the system of equations above?

A. There are exactly 4 solutions.

B. There are exactly 2 solutions.

C. There is exactly 1 solution.

D. There are no solutions.

2.24 fcee5450-b85d-43d4-a5c2-befadc67d6b0 · Hard

$$57x^2 + (57b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A. $\frac{1}{57}$

B. $\frac{1}{19}$

C. 1

D. 57

2.25 b329914a-0391-4290-8ade-dc32575f2627 · Hard

$$\sqrt{k - x} = 58 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Student-produced response: _____

2.26 f7813f1c-5e31-444e-8273-c1fc41ce7f59 · Hard

$$y + k = x + 26$$

$$y - k = x^2 - 5x$$

In the given system of equations, k is a constant. The system has exactly one distinct real solution. What is the value of k?

Student-produced response: _____

2.27 6c2521be-efeb-4288-b874-77e68fac25dd · Hard

$$5x^2 + 10x + 16 = 0$$

How many distinct real solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.28 2ad6a2c3-2e30-4705-883b-851e5cc23933 · Hard

$$-5x + 13 = 73$$

What is the sum of the solutions to the given equation?

A. $-\frac{146}{5}$

B. -12

C. 0

D. $\frac{26}{5}$

2.29 c9ba957d-0aef-4fba-942a-16b7f4d4d31b · Hard

$$24 - x + 34 - x = 25$$

What is the positive solution to the given equation?

Student-produced response: _____

2.30 22d60924-a9af-42ee-8e63-30a79298bfb3 · Hard

$$-16x^2 - 8x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c?

Student-produced response: _____

2.31 75273fff-e1a3-4a75-a92d-7cecf27c2bd7 · Hard

$$xkx - 56 = -16$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k?

Student-produced response: _____

2.32 6b9ef50f-0c40-4a0e-bd2f-1b39517b56d6 · Hard

$$y = 18$$

$$y = -3x - 18^2 + 15$$

If the given equations are graphed in the xy-plane, at how many points do the graphs of the equations intersect?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.33 ba1a8be9-37d7-4a8f-80c3-300f4e67565d · Hard

$$y = x + 9$$

$$y = x^2 + 16x + 63$$

A solution to the given system of equations is x, y. What is the greatest possible value of x?

A. -6

B. 7

C. 9

D. 63

2.34 ccdcaf2a-7d93-4de9-b26e-676ccc65d3d4 · Hard

$$4x^2 - 5x - 5 = 0$$

What is the greatest solution to the given equation?

A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$

B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$

C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$

D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

2.35 6bc7b975-d9d8-4ebf-a201-229c5335d826 · Hard

$$64x^2 - 16a + 4bx + ab = 0$$

In the given equation, a and b are positive constants. The sum of the solutions to the given equation is $k4a + b$, where k is a constant. What is the value of k?

Student-produced response: _____

2.36 c7ec775c-6d72-46ae-8501-bb2d64052015 · Hard

$$y = -2.5$$

$$y = x^2 + 8x + k$$

In the given system of equations, k is a positive integer constant. The system has no real solutions. What is the least possible value of k?

Student-produced response: _____

2.37 016807-DC · Hard

$2x^2 - 2 = 2x + 3$

Which of the following is a solution to the equation above?

A. 2

B. $1 - \sqrt{11}$

C. $\frac{1}{2} + \sqrt{11}$

D. $\frac{1 + \sqrt{11}}{2}$

2.38 08326-DC · Hard

$2x^2 - 4x = t$

In the equation above, t is a constant. If the equation has no real solutions, which of the following could be the value of t ?

A. -3

B. -1

C. 1

D. 3

2.39 ab551853-877a-4560-a1dc-8c5843c3aa42 · Hard

$$8x + y = -11$$

$$2x^2 = y + 341$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy-plane. What is a possible value of x?

A. -15

B. -11

C. 2

D. 8

2.40 5f4a4830-04bb-473f-ac93-a66d45672ea3 · Hard

$$4x - 3 = -9$$

How many solutions does the given equation have?

A. Zero

B. One

C. Two

D. Infinitely many

2.41 00272e26-db4e-4b57-b229-9040d402e42f · Hard

$$y = -1.5$$

$$y = x^2 + 8x + a$$

In the given system of equations, a is a positive constant. The system has exactly one distinct real solution. What is the value of a?

Student-produced response: _____

2.42 b026d255-ee45-4659-b390-1b694d47296b · Hard

$$\frac{14x}{7y} = 2\sqrt{w + 19}$$

The given equation relates the distinct positive real numbers w, x, and y. Which equation correctly expresses w in terms of x and y?

A. $w = \sqrt{\frac{x}{y}} - 19$

B. $w = \sqrt{\frac{28x}{14y}} - 19$

C. $w = \frac{x^2}{y} - 19$

D. $w = \frac{28x^2}{14y} - 19$

2.43 51befb2a-39f3-4194-bd65-5c08a93e42cc · Hard

$$x^2 + 7x + 5 = 0$$

One solution to the given equation can be written as $x = \frac{-7 + \sqrt{k}}{2}$, where k is a constant. What is the value of k?

Student-produced response: _____

2.44 7e4cd51c-5329-4e8b-83dd-1416f04a294f · Hard

$$x^2 + y + 7 = 7$$

$$20x + 100 - y = 0$$

The solution to the given system of equations is x, y. What is the value of x?

Student-produced response: _____

2.45 11192-DC · Hard

$\sqrt{2x+6}+4=x+3$

What is the solution set of the equation above?

A. $\{-1\}$

B. $\{5\}$

C. $\{-1, 5\}$

D. $\{0, -1, 5\}$

2.46 91e78971-fbd4-44f8-ad04-97fb96cf9931 · Hard

$$y = 2x^2 - 21x + 64$$

$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x?

A. -8

B. -6

C. 6

D. 8

2.47 87a33c1a-0dd6-4c06-953e-5ebde0abb83e · Hard

$$-2x^2 + 20x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c?

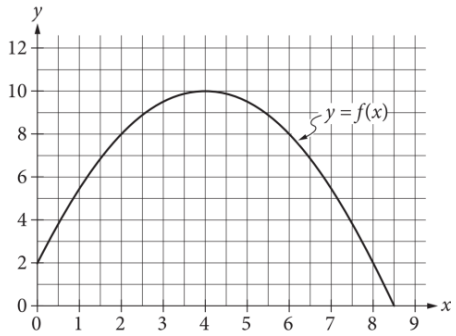
A. -68

B. -50

C. -32

D. 0

2.48 070621-DC · Hard



The graph of the function f, defined by $f(x) = -\frac{1}{2}(x-4)^2+10$, is shown in the xy-plane above. If the function g (not shown) is defined by $g(x) = -x+10$, what is one possible value of a such that $f(a) = g(a)$?

Student-produced response: _____

2.49 52397fe6-3bd5-4193-935e-114ea31cb4b0 · Hard

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

In the given equation, c is a positive constant. Which of the following is one of the solutions to the given equation?

A. $-c$

B. $-c^2 - 39^2$

C. $-\sqrt{39^2 - c^2}$

D. $-\sqrt{c^2 + 39^2}$

2.50 2ce9766f-6e56-4ce6-be93-f8ca5401d22d · Hard

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p, q, r, and s. Which of the following is equivalent to q?

A. $p + r + s$

B. $20p + r + s$

C. $\frac{prs}{pr + ps + rs}$

D. $\frac{prs}{20p + 20r + 20s}$

2.51 8204e6e4-4fd9-46f0-8199-646409cd8719 · Hard

$$x - 47^2 = 1$$

What is the sum of the solutions to the given equation?

Student-produced response: _____

2.52 06323-DC · Hard

In the xy-plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y = x$ at the points $(0, 0)$ and (a, a). What is the value of a ?

Student-produced response: _____

2.53 05327-DC · Hard

$$x - y = 1$$
$$x + y = x^2 - 3$$

Which ordered pair is a solution to the system of equations above?

A. $(1 + \sqrt{3}, \sqrt{3})$

B. $(\sqrt{3}, -\sqrt{3})$

C. $(1 + \sqrt{5}, \sqrt{5})$

D. $(\sqrt{5}, -1 + \sqrt{5})$

2.54 029000-DC · Hard

$$D = T - \frac{9}{25}(100 - H)$$

The formula above can be used to approximate the dew point D, in degrees Fahrenheit, given the temperature T, in degrees Fahrenheit, and the relative humidity of H percent, where $H > 50$. Which of the following expresses the relative humidity in terms of the temperature and the dew point?

A. $H = \frac{25}{9}(D - T) + 100$

B. $H = \frac{25}{9}(D - T) - 100$

C. $H = \frac{25}{9}(D + T) + 100$

D. $H = \frac{25}{9}(D + T) - 100$

2.55 47af8694-05ad-4849-a1fb-69bbee9cc524 · Hard

$$64x^2 + bx + 25 = 0$$

In the given equation, b is a constant. For which of the following values of b will the equation have more than one real solution?

A. -91

B. -80

C. 5

D. 40

2.56 c442bd01-c336-457a-a300-cc899742f16e · Hard

$$5(x + 7) = 15x - 17x + 7$$

What is the sum of the solutions to the given equation?

Student-produced response: _____

2.57 c324ef1d-80fd-41a5-9609-8658df690543 · Hard

$$x - 1^2 = -4$$

How many distinct real solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.58 cf311c5e-be31-4641-9700-9b164150cb1e · Hard

Which quadratic equation has no real solutions?

A. $x^2 + 14x - 49 = 0$

B. $x^2 - 14x + 49 = 0$

C. $5x^2 - 14x - 49 = 0$

D. $5x^2 - 14x + 49 = 0$

2.59 30e5c8c1-d494-443c-a28f-3d8867feefed · Hard

$$xx + 1 - 56 = 4xx - 7$$

What is the sum of the solutions to the given equation?

Student-produced response: _____

2.60 031698-DC · Hard

In the xy-plane, the graph of $y = x^2 - 9$ intersects line p at $(1, a)$ and $(5, b)$, where a and b are constants. What is the slope of line p ?

A. 6

B. 2

C. -2

D. -6

2.61 7f98a663-e37c-478d-90b1-a6bf6fc13f6b · Hard

$$\frac{1}{7}x - k16 - x^2 = 0$$

In the given equation, k is a positive constant. The sum of the solutions to the equation is 59. What is the value of k?

Student-produced response: _____

2.62 e0fc5fe8-0ada-49b3-935b-b2eea9b98cff · Hard

$$r^2 + qr = 2r - 55$$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q?

Student-produced response: _____

2.63 2dabc1f7-ef1b-4948-88ab-4a73835ddf68 · Hard

In the xy-plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c?

A. $-\frac{481}{4}$

B. -100

C. $-\frac{319}{4}$

D. $-\frac{9}{2}$

2.64 11289-DC · Hard

If $u-3=\frac{6}{t-2}$, what is t in terms of u ?

A. $t=\frac{1}{u}$

B. $t=\frac{2u+9}{u}$

C. $t=\frac{1}{u-3}$

D. $t=\frac{2u}{u-3}$

2.65 28f04aa6-9eca-4e86-bee8-86e106dc02df · Hard

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k?

A. 8

B. 10

C. 20

D. 40

2.66 78178dc9-f607-45b8-919f-337b38be824a · Hard

$$\frac{1}{4xy} + xyz = \frac{1}{3yz}$$

In the given equation, x, y, and z are positive numbers. Which expression is equivalent to y?

A. $\frac{4x - 3z}{12x^2 z^2}$

B. $\sqrt{\frac{4x - 3z}{12x^2 z^2}}$

C. $\frac{1}{3xz^2 - 4x^2 z}$

D. $\sqrt{\frac{1}{3xz^2 - 4x^2 z}}$

2.67 3a6913de-0959-40fb-980c-a37f8b9ee578 · Hard

$$\frac{12}{n} - \frac{2}{t} = -\frac{2}{w}$$

The given equation relates the variables n, t, and w, where $n > 0$, $t > 0$, and $w > t$. Which expression is equivalent to n?

A. $12tw$

B. $6t - w$

C. $\frac{w - t}{6tw}$

D. $\frac{6tw}{w - t}$

2.68 4d5ada3e-eaea-405e-9b02-b7797ddf34f3 · Hard

If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?

Student-produced response: _____

2.69 b40bb858-a515-41ac-963c-90f9040a57b8 · Hard

$$x - 9 + 45 = 63$$

What is the sum of the solutions to the given equation?

Student-produced response: _____

2.70 441e80ee-d059-4eb1-b33c-69ae14eabe0e · Hard

$$\sqrt{x - 2^2} = \sqrt{3x + 34}$$

What is the smallest solution to the given equation?

Student-produced response: _____

2.71 207c18b2-0202-4869-bc5f-64c10fac4239 · Hard

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point $-5, y$, where $y < 0$. Which expression represents the value of b?

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

2.72 2c7c5018-d050-4b02-9835-da06930bf5a0 · Hard

The solutions to $x^2 + 6x + 7 = 0$ are r and s, where $r < s$. The solutions to $x^2 + 8x + 8 = 0$ are t and u, where $t < u$. The solutions to $x^2 + 14x + c = 0$, where c is a constant, are $r + t$ and $s + u$. What is the value of c?

Student-produced response: _____

2.73 09289-DC · Hard

$$y = x^2 + 2x + 1$$
$$x + y + 1 = 0$$

If (x_1, y_1) and (x_2, y_2) are the two solutions to the system of equations above, what is the value of $y_1 + y_2$?

A. -3

B. -2

C. -1

D. 1

2.74 c9d1415e-a7ee-4cc6-826b-fc650c58fc3b · Hard

$$x^2 - 34x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n?

Student-produced response: _____

2.75 05702-DC · Hard

During a 5-second time interval, the average acceleration a, in meters per second squared, of an object with an initial velocity of 12 meters per second is defined by the equation $a = \frac{v_f - 12}{5}$, where v_f is the final velocity of the object in meters per second. If the equation is rewritten in the form $v_f = xa + y$, where x and y are constants, what is the value of x ?

Student-produced response: _____

2.76 6dc78b73-3d3a-47a2-8b1b-c6041a6a7337 · Hard

In the *xy*-plane, a line with equation $2y = c$ for some constant c intersects a parabola at exactly one point. If the parabola has equation $y = -2x^2 + 9x$, what is the value of c?

Student-produced response: _____

2.77 11064-DC · Hard

$$\frac{1}{x^2+10x+25} = 4$$

If x is a solution to the given equation, which of the following is a possible value of $x+5$?

A. $\frac{1}{2}$

B. $\frac{5}{2}$

C. $\frac{9}{2}$

D. $\frac{11}{2}$

2.78 573f8b0a-9818-4498-97d9-46bab82da00c · Hard

$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as $\frac{8 - \sqrt{k}}{4}$, where k is a constant. What is the value of k?

Student-produced response: _____

2.79 3855194f-20fa-4907-9dfd-8097f615bdff · Hard

$$x - 29 = x - ax - 29$$

Which of the following are solutions to the given equation, where a is a constant and $a > 30$?

I. a
II. $a + 1$
III. 29

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

2.80 8acbff18-5c28-43c6-85f6-a6d5078b92cd · Hard

In the xy-plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b?

Student-produced response: _____

2.81 025616-DC · Hard

$$\frac{4x^2}{x^2-9} - \frac{2x}{x+3} = \frac{1}{x-3}$$

What value of x satisfies the equation above?

A. -3

B. $-\frac{1}{2}$

C. $\frac{1}{2}$

D. 3

2.82 5069a621-921f-40db-8e07-7253b4a9383b · Hard

$$x^2 - 40x - 10 = 0$$

What is the sum of the solutions to the given equation?

A. 0

B. 5

C. 10

D. 40